

# Yan Yan

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## ACADEMIC REFERENCE

**Professor Salvador Pané**, Multi-Scale Robotics Lab, ETH Zurich [vidalp@ethz.ch](mailto:vidalp@ethz.ch)  
**Dr. Luis Deán-Ben**, Group Leader, Multiscale Functional and Molecular Imaging Lab, ETH&UZH [xl.deanben@pharma.uzh.ch](mailto:xl.deanben@pharma.uzh.ch)

## EDUCATION

**M.Sc., Biomedical Engineering, University Basel, Switzerland** 09.2023-Present  
**Current GPA:** 5.3/6

**Main Courses:** Principles of Medical Imaging, Advanced Methods in Medical Image Analysis, Model-Based Signal Processing, Optimization Methods, LTI Systems and Control, Mechanics, Modelling & Simulation (COMSOL), Materials Science and Biomaterials, Electrical Fundamentals

**B.Sc., Dental Technology, Sichuan University, West China School of Stomatology, China** 09.2019-06.2023  
**GPA:** 88.24/100

**Main Courses:** Mathematical Analysis, Advanced Algebra, Analytic Geometry, Medical Statistics, Medical Physics, Medical Biology, Histology and Pathology, Anatomy and Physiology, Organic Chemistry and Experiments, Dental Materials, Oral and Maxillofacial Imaging Diagnostics, Dental Prosthesis Fabrication

## RESEARCH INTERNSHIP

**Multiscale Functional and Molecular Imaging Lab, ETH & UZH, Switzerland (Master Thesis)** 07.2025-Present  
**Advisor:** Dr. Luis Deán-Ben

- Performed optoacoustic image reconstruction using homogeneous and heterogeneous back-projection as well as LSQR-based heterogeneous models to reconstruct in vivo mouse abdomen images
- Implemented and compared vesselness enhancement filters (Frangi, Jerman, and others) for post-processing
- Developed an unrolled, physics-guided deep learning framework that integrates CNN-based image enhancement with model-based reconstruction, where the CNN acts as a self-learned regularizer
- Conducted systematic comparison and evaluation of reconstruction performance across different algorithm

**Multi-Scale Robotics Lab, ETH Zurich, Switzerland (Semester Thesis)** 10.2024-07.2025  
**Advisor:** Huimin Chen, Dr. Semih Sevim, (under supervision of Prof. Salvador Pané)  
**Grade:** 5.75/6

- Determined and optimized PEGDA/PNIPAM hydrogel formulations for DLP-based 3D printing
- Fabricated PDMS-acrylic pool chambers for hydrogel photocuring via PDMS spin-coating and oxygen-plasma bonding
- Designed and 3D-printed bilayer hydrogel cuff electrodes integrating PNIPAM and PEGDA
- Controlled curvature deformation under thermal stimulation through structural design optimization
- Designed and 3D-printed flow chambers to evaluate cuff-electrode adhesion under fluid conditions
- Utilized CAD modeling in Onshape, UV polymerization setup, and optical imaging techniques for characterization

## INDUSTRIAL INTERNSHIP

**Chengdu DT Denture Technology Development Co., Ltd., China** 02.2023-06.2023  

- Learned the management mode of the company in the process of denture production
- Designed and fabricated the dental prosthesis via CAD/CAM

**West China Hospital of Stomatology, Sichuan University, National Center of Stomatology** 07.2022-06.2023  

- Assisted clinical doctors to complete patients' diagnoses, information collection and medical scanning
- Mastered the use of oral scan, facial scan and electronic facebow
- Designed and fabricated the dental prosthesis via CAD/CAM

**Chengdu Boltzmann Zhibei Intelligence Technology Co., Ltd., China** 06.2021-09.2021  

- Referred to the literature about the CBCT images quality and airway analysis on lateral cephalogram
- Processed the CBCT images via the ITK-SNAP to segment and mark the dentition
- Participated in a research-oriented project related to cervical vertebral maturation analysis

## PUBLICATION

Jiang F, Abdulqader AA, Yan Y, Cheng F, Xiang T, Yu J, Li J, Qiu Y, Chen X. Deep learning based quantitative cervical vertebral maturation analysis. [Accepted by Head Face Med.2025]  

- Contributed to manuscript preparation and assisted in data summary and result interpretation

## SKILLS

**Language:** Mandarin (native), English (fluent), German (basic)  
**Programming:** Python, MATLAB, LaTeX, R, HTML/CSS **Hobby:** Reading, Traveling, Swimming, Wandering